

Scientific Thinking: From Claims to Evidence

A course architecture that moves students from questioning claims to constructing evidence-based scientific arguments.

Natalia Borrego | Adapted from Scientific Thinking courses taught at the American University in Cairo, including the Fall 2020 course sequence.

Design principle. Students repeatedly practice asking what a claim means, what evidence would distinguish among explanations, how research design affects inference, and how conclusions should change when the evidence changes.

COURSE PROGRESSION

STAGE 01 Question • Science and non-science • Skepticism, empiricism, rationalism • Hypothesis formulation • Falsifiability	STAGE 02 Test • Research ethics • Internal, external, construct, statistical validity • Research methods • Correlation and causation	STAGE 03 Evaluate • Bias and assumptions • Peer-reviewed literature • Induction, deduction, abduction • Validity and soundness	STAGE 04 Create • Select a scientific topic • Find primary literature • Develop a case study and questions • Revise after feedback
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BLOOM'S TAXONOMY AS AN INSTRUCTIONAL SCAFFOLD

UNDERSTAND concepts and claims	APPLY reasoning to new cases	ANALYZE methods and evidence	EVALUATE competing arguments	CREATE an evidence-based product
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EARLY-COURSE CASE

Clever Hans

Students ask what would count as evidence, generate alternative explanations, and identify tests that could discriminate among them.

HYPOTHESIS PRACTICE

Beri-beri case study

Students turn observations into falsifiable hypotheses rather than treating a plausible explanation as a conclusion.

INFERENCE PRACTICE

Correlation or causation?

Students decide what the evidence can support, which variables remain uncontrolled, and what additional tests would strengthen inference.

LITERATURE PRACTICE

Evaluating scientific papers

Students identify research questions, evidence, assumptions, and limits of inference before using literature in their own final project.

Practice precedes evaluation; feedback precedes the final product.

The course sequence mixed low-stakes quizzes and short case studies with larger assignments and an iterative group project. Students first practiced individual reasoning skills, then combined them when evaluating literature and constructing their own case study.

Assessment form	Role in the course	What students practice
Low-stakes quizzes and modules	Frequent checks before or after class activities.	Terminology, conceptual foundations, and retrieval needed for higher-order work.
Short case studies	Apply reasoning to unfamiliar claims and scenarios.	Hypothesis testing, identifying assumptions, and distinguishing evidence from explanation.
Major homework	Integrate several skills at once.	Experimental design, logical reasoning, quantitative reasoning, and scientific argument.
Iterative term project	Select a topic, locate peer-reviewed literature, develop an outline, submit a section for feedback, revise, and complete the final case study.	Independent literature use, synthesis, question construction, revision, and communication.

What this design makes visible. By the end of the course, students are expected not only to recognize the vocabulary of scientific reasoning, but to use it: decide what a claim predicts, evaluate whether a test is valid, interpret evidence cautiously, and construct an argument that can be challenged and revised.

EXAMPLE SEMESTER ARC

Weeks 1-4 | Nature of science

Scientific thinking, science vs. pseudoscience, skepticism, empiricism, rationalism, falsifiability, hypothesis formulation.

Weeks 5-9 | Research and evidence

Ethics, research validity, methods, data interpretation, correlation and causation, visual representation of data.

Weeks 10-13 | Evaluating arguments

Biases and assumptions, peer-reviewed literature, logical reasoning, induction/deduction/abduction, validity and soundness.

Weeks 14-15 | Creating and revising

Literature synthesis, case-study construction, question development, section-level feedback, revision, and final submission.

Course endpoint. The final project asks students to combine the course's components rather than complete one more isolated exercise: locate evidence, evaluate it, construct a case, develop questions, receive feedback, and revise.